

1. Administrative & Study Identification

Full Study Title: A Phase 1/2 Open-Label, Umbrella Platform Design Study of Investigational Agents in Combination With Pembrolizumab (MK-3475) With or Without Chemotherapy in Participants With 1L Locally Advanced Unresectable/Metastatic Esophageal Cancer: KEYMAKER-U06 Substudy 06E **Short Title/Acronym:** Substudy 06E: Umbrella Study of Combination Therapies in Esophageal Cancer (MK-3475-06E/KEYMAKER-U06) **Protocol Number:** MK-3475-06E **NCT Number:** NCT06780111 **Posting ID:** 440358557860943556 **Sponsor Name:** Merck **Investigational Product:** Pembrolizumab (MK-3475), Ifinatumab deruxtecan (I-DXd), Sacituzumab tirumotecan, Oxaliplatin, Fluorouracil (5-FU; Trade names: Carac, Efudex, Adrucil, Fluoroplex, Tolak), Leucovorin **Phase of Development:** Phase 1 / Phase 2 **Study Status:** Recruiting **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of investigational agents (including I-DXd and sacituzumab tirumotecan) combined with pembrolizumab with or without chemotherapy, determine the recommended phase 2 dose (RP2D) of I-DXd, and measure specific safety outcomes including Dose Limiting Toxicities (DLTs) and adverse event frequencies. **Secondary Objectives:** To evaluate Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), and objective response rate (ORR).

2.2 Background & Rationale There is a substantial clinical need for more effective and tolerable first-line (1L) treatment options for patients with advanced esophageal squamous cell carcinoma (ESCC). B7-H3 is a transmembrane protein highly expressed in several cancers, including ESCC, and is associated with a poor prognosis. Ifinatumab deruxtecan (I-DXd) is a B7-H3-directed antibody-drug conjugate (ADC) that has previously shown promising anti-tumor activity in ESCC as monotherapy. This KEYMAKER-U06 umbrella substudy evaluates these combinations to improve 1L outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Platform Study) **Control Type:** Active Comparator (Arm 1: Pembrolizumab + mFOLFOX6) **Blinding:** Open-label **Randomization:**

Randomized (1:2 to the reference arm and investigational arms in the randomized phase)

3.2 Planned Enrollment & Duration Target **\$N\$:** 228 participants
Number of Sites: Global Multicenter **Estimated Study Start:** 30-Jul-2025 **Estimated Primary Completion:** 04-Jan-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of locally advanced unresectable or metastatic squamous cell carcinoma of the esophagus (ESCC) in the first-line (1L) setting. 3. Measurable disease per RECIST 1.1 as assessed by the local site investigator/radiology and verified by BICR. Lesions situated in a previously irradiated area are considered measurable if progression has been shown. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 5. Has adverse events (AEs) due to previous anticancer therapies of $\leq$ Grade 1 or baseline (except alopecia and vitiligo). Endocrine-related AEs adequately treated are acceptable. 6. HIV-infected participants must have well-controlled HIV on antiretroviral therapy (ART). 7. Hepatitis B (HBsAg positive) participants are eligible if on antiviral therapy ≥ 4 weeks with undetectable viral load. 8. Hepatitis C (HCV) participants are eligible if HCV viral load is undetectable.

4.2 Exclusion Criteria 1. Has had prior systemic anticancer therapy for locally advanced unresectable or metastatic esophageal cancer. 2. Tumor invasion into adjacent organs (e.g., aorta or respiratory tract) at increased risk of fistula. 3. Uncontrollable pleural effusion, pericardial effusion, or ascites requiring frequent drainage/intervention. 4. Clinically significant corneal disease. 5. Prior therapy with anti-PD-1, anti-PD-L1, anti-PD-L2, B7-H3-targeted agents (e.g., orlotamab, enoblituzumab), or topoisomerase-I inhibitor (including ADCs). 6. Prior systemic anticancer therapy within 4 weeks, or prior radiotherapy within 2 weeks of first dose. 7. Received a live or live-attenuated vaccine within 30 days before the first dose. 8. Inadequate cardiac function (QTcF > 470 msec) or clinically significant cardiovascular disease within 6 months. 9. Peripheral neuropathy $\geq$ Grade 2. 10. Diagnosis of immunodeficiency or receiving systemic steroids (>10 mg daily prednisone equivalent) within 7 days prior. 11. Known additional malignancy progressing or requiring active treatment within the past 3 years. 12. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 13. Active autoimmune disease requiring systemic treatment in the past 2 years. 14. History of (noninfectious) pneumonitis/interstitial lung disease (ILD) that required steroids, or current/suspected pneumonitis/ILD. 15. Active infection requiring systemic therapy or severe pulmonary compromise. 16. Severe hypersensitivity ($\geq$ Grade 3) to mAbs,

pembrolizumab, I-DXd, chemotherapy agents/excipients, murine proteins, or platinum. 17. Prior allogeneic tissue/solid organ transplant. 18. Inadequate recovery from major surgery or incapacitated.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple study arms evaluating combinations: * **Arm 1 (Reference):** Pembrolizumab 200 mg Q3W + mFOLFOX6 (oxaliplatin 85 mg/m² Q2W + fluorouracil 400 mg/m² bolus and 2400 mg/m² continuous Q2W + leucovorin 400 mg/m² Q2W). * **Investigational Arms:** Include Pembrolizumab (200 mg Q3W or 400 mg Q6W) in combination with I-DXd (8 or 12 mg/kg Q3W) or Sacituzumab tirumotecan (4 mg/kg Q6W), with or without fluorouracil and oxaliplatin. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Pembrolizumab for up to 18 cycles (Q6W) or 35 cycles (Q3W) equating to approximately 2 years; other agents administered until progressive disease (PD) or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications. **Prohibited:** Other investigational therapeutic agents, systemic treatment for autoimmune diseases, and systemic steroids for pneumonitis/ILD.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Imaging, RECIST v1.1 Assessment
Baseline	Day 0	Randomization, First Dose of Study Interventions
Interim	Q2W, Q3W, or Q6W	Safety Labs, Efficacy Assessment, Infusion administration corresponding to specific arm assignments
End of Study	Follow-Up	Treatment discontinuation, final safety assessments, Overall Survival tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- Percentage of Participants who Experience Dose Limiting Toxicities (DLTs) During the Safety Lead-In Phase:** Evaluated at approximately 21 days post-first dose.
- Percentage of Participants who Experience an Adverse Event (AE):** Measured throughout the study duration.
- Percentage of Participants Who Discontinue Study Intervention Due to an AE:** Measured throughout the study duration.

7.2 Secondary & Exploratory Endpoints

- * Duration of Response (DOR).
- * Progression-Free Survival (PFS).
- * Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard monitoring spanning throughout the treatment; a dedicated safety lead-in phase utilizes a Bayesian optimal interval design to confirm the safety and DLTs of combinations. **Serious Adverse Events (SAEs):** Reported per standard oncology protocols (typically within 24 hours). **Data Monitoring Committee (DMC):** Internal and external safety review governance structures implemented by the Sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy outcomes; Safety Set for AE/tolerability evaluation. **Sample Size Justification:** Approximately 228 enrolled participants; utilizing a safety lead-in followed by a randomized phase to confidently detect efficacy and ORR shifts. **Handling of Missing Data:** Appropriate right-censoring methods for time-to-event outcomes (OS, PFS, DOR).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard clinical site monitoring with specialized oversight during the Bayesian safety lead-in phase. **Audit Trail:** eCRF documentation along with Blinded Independent Central Review (BICR) for all crucial imaging data.

11. Study Identifiers & Governance

Primary ID: {{MK-3475-06E}} **Posting ID:** {{440358557860943556}}
Registry Number: {{NCT06780111}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{Substudy 06E: Umbrella Study of Combination Therapies in Esophageal Cancer (MK-3475-06E/KEYMAKER-U06)}} **Official Regulatory Title:** {{A Phase 1/2 Open-Label, Umbrella Platform Design Study of Investigational Agents in Combination With Pembrolizumab (MK-3475) With or Without Chemotherapy in Participants With 1L Locally Advanced Unresectable/Metastatic Esophageal Cancer: KEYMAKER-U06 Substudy 06E}}

12. Clinical Framework (Oncology Specific)

Primary Condition: {{Esophageal Squamous Cell Carcinoma (ESCC)}}
Diagnosis Threshold: {{Locally advanced unresectable or metastatic ESCC}} **Medical Methodology:** {{Umbrella trial of Immune

Checkpoint Inhibitor combined with ADCs and/or Chemotherapy}}

Optimization Target: {{Objective Response Rate (ORR) and Safety/ RP2D confirmation}}

13. Study Procedures & Methodology

Management Pathway: {{KEYMAKER Umbrella Platform Protocol}}

Education Protocol (HCP): {{Safety management criteria for Antibody-Drug Conjugates and IO combination toxicities}}

Education Protocol (Patient): {{Onsite informed consent and training on reporting Immune-Related Adverse Events (irAEs)}}

Quality Audit Mechanism: {{Blinded Independent Central Review (BICR) for RECIST v1.1 imaging assessments}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 35 cycles (~2 years) plus subsequent overall survival follow-up}}

Onsite Visit Cadence: {{Every 2 to 6 weeks, aligned with Q2W, Q3W, or Q6W IV infusion schedules}}

Remote Monitoring Frequency: {{Routine tracking during post-treatment follow-up phase}}

Checklist Review Interval: {{Prior to each treatment cycle}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				